

Fault rate analysis, modelling and estimation

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This paper describes the development and application of a generic fault generation model applicable to copper and other network technologies. This model is intended to act as a framework for understanding, analysing and predicting the impact of existing and new initiatives on the fault generation process. The paper includes an overview of the approach and the tools, such as regression analysis, which are used when applying the model and shows the results from two case studies which look at fault generation at a zonal level, and fault generation during a period of network improvement. These studies suggest that the main fault generation mechanisms are manual intervention and unusual weather conditions, and show that improvement programmes are likely to introduce a temporary increase in fault rates during the improvement period before the benefits are felt. For this reason, it is suggested that the introduction of any new processes (i.e. working practices), tools (i.e. fault location), or technologies should therefore minimise the impact of intervention and be able to withstand adverse environmental conditions, and that any such programmes should be designed to involve minimal manual intervention.

1. Introduction

Forecasts should be seen as ‘guideposts against which future developments may be evaluated’ — Bank of England.

The overall fault volume within BT’s local access network has remained fairly constant over a number of years despite a steady increase in the total number of exchange connections. As a result, the number of faults per exchange connection (faults/EC) has steadily fallen over this period, apart from the ‘storm’ years of 1987 and 1989 (financial year end 1988 and 1990), as illustrated in Fig 1. However, with increased competition and a desire to add value for shareholders and customers, there is a continual pressure to reduce fault rates. This, therefore, presents a challenge as to how further reductions can be achieved and sustained.

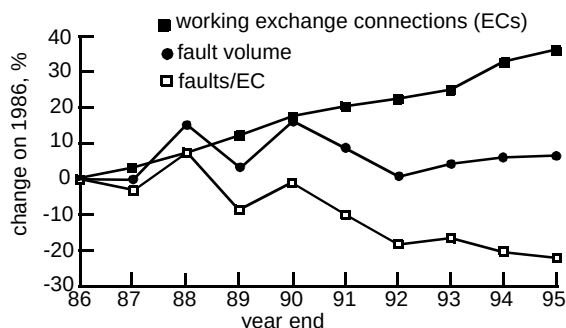


Fig 1 Change in fault rates over 10 years.

As part of the existing improvement process, a number of programmes and initiatives are being carried out within the local access BT copper network which have an impact on the fault generation process. These programmes are, in general, aiming to ‘enhance customer satisfaction by increasing the reliability of the network’ in line with company objectives, while reducing costs where possible.

To understand the effects of both existing and new programmes, there is therefore a need to model this fault generation process to enable a number of requirements to be met. These requirements may include:

- quantifying financial benefits of new and existing programmes,
- assessing future fault trends to assess impact on resourcing,
- designing new programmes which will have minimal impact on fault generation.

In addition to these general requirements, there was also a need to define a generic model or ‘framework’ to accommodate differing viewpoints as to why the network fails. This ‘framework’ can be used to improve co-ordination between groups operating within the access network by providing a common basis of understanding and a focus for future effort.

The aim of the work was therefore to develop a generic fault generation model that could be applied both to the

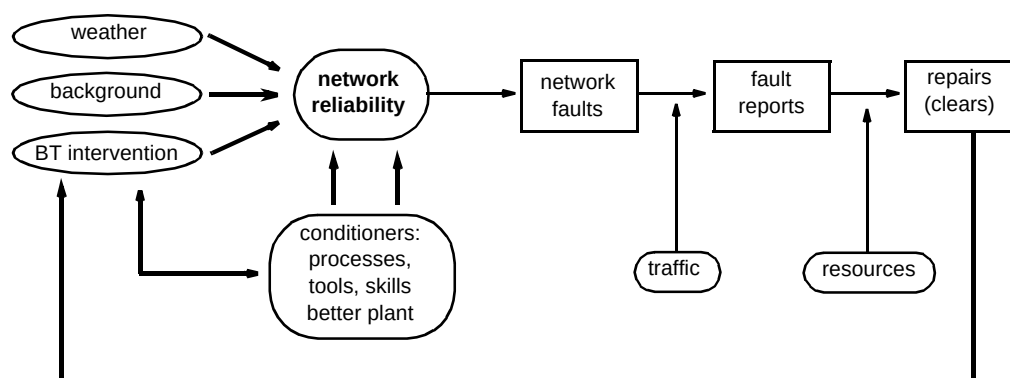


Fig 2 Access network generic fault model.

copper access network and to other technologies (such as fibre) for evaluating the impact of current and future investment programmes which may influence plant ‘whole-life’ costs [1]. This paper describes the development and application of a generic fault generation model applicable to the BT network. An overview is included of the approach and the tools, such as regression analysis, which were used when applying the model. The application of the model and analysis methods is then demonstrated by using two case studies which look at fault generation for an average zone, and at fault generation at exchange level during a period of network improvement.

2. Model

The generic reliability model, as illustrated in Fig 2, was therefore built using expertise on network behaviour and combined with traditional reliability theory. For example, the theoretical approach can categorise failure causes into [2]:

- intrinsic stress (primary failure) such as natural ageing, design, manufacturing or installation defects,
- excessive stress (secondary failure) due to environment, personnel or other equipment,
- control errors (command failure).

This model deals with environment (as weather), personnel (as BT intervention), and intrinsic failures (as background) and are considered as fault drivers (i.e. faults follow shortly after intervention or weather events). Both weather and intervention are often quoted as reasons for network failures (although usually not quantified) and are therefore used as the drivers within the model to show consistency with operating experience. Command failures are not considered to be significant in the proposed model due to the straightforward nature of the technology employed within the analogue access network.

While the modelling of drivers is fairly straightforward (i.e. stressing equipment leads to faults), the effects of plant reliability changing over time is not. Many of the theoretical

approaches assume that the distribution profile of ‘strength’ (or reliability) against component age is fixed [1—3]. Also, these distributions are often considered for a large population of a single type of component (i.e. a block terminal). However, within the access network, the reliability profile of the plant changes continually due to factors such as new plant (i.e. technology), changes in working practices (i.e. processes), and new maintenance tools (i.e. diagnostics and fault location). Therefore, this changing ‘reliability’ has been incorporated into the model by including a ‘conditioning’ input to account for the changes in technology, tools, and processes.

To complete the model of the fault-generation process, the fault detection and clear processes (i.e. fault handling) are also included. A significant fault-finding mechanism is through customers making calls, and the ability to clear the faults is dependent on the resources available. This loop is important when considering that the repair process forms part of the network intervention process.

3. Approach

One approach to building a mathematical model is to understand the variables and physical relationships that affect the behaviour of whatever is being analysed, the alternative being to use a ‘black-box’ approach where the internal transfer mechanisms are ignored (i.e. fundamental versus technical analysis [4]). An advantage of understanding the transfer mechanisms is that a model is less likely to incorporate input data which is not relevant, but might have coincidental relationships with the effects. In addition, the effects of updating the model to incorporate changes can be more readily assessed when changes are introduced into the actual system (which is inevitable).

A number of approaches to the analysis were available due to the large number of data sources (e.g. financial, manpower) at varying levels of granularity (e.g. by year, by month, by exchange, by ‘zone’, nationally). In practice, a top-down approach was used starting at the most general level (i.e. nationally, by year), with increasing granularity occurring as the research progressed. Use was made of the

data that was readily available, and, where possible, being as near to the physical fault generation process as possible (i.e. manpower rather than financial data).

Initially, yearly financial data was analysed at a national level to assess the impact of expenditure by various improvement programmes on the fault rate. The results obtained from this provided an initial view as to the net benefits by programme (i.e. final benefits obtained less the faults caused during improvements), but lacked sufficient detail or granularity. For this reason, more detailed analysis was carried out using man-hour data over monthly time periods both at national, zonal, and exchange level (where possible) [5, 6]. It is this more detailed analysis which is presented in this paper. Other research at the exchange level for use on a day-to-day basis (operational) is also being pursued, but is outside the scope of this paper.

4. Methods

A number of techniques are available for finding the optimum combination of input data which provide the best 'fit' to a set of output data. In all cases, whether implicitly or explicitly, the techniques use a 'least squares' error minimisation technique. This section is intended to provide an overview of the techniques available for use in analysing the case studies presented in this paper. These techniques include:

- manual adjustment to provide the best visual fit,
- error minimisation by using numerical solving algorithms,
- regression analysis,
- neural networks.

For the fault modelling presented in this paper, visual checks have been used first, followed by simple error minimisation, and then extended to include regression analysis for more qualitative solutions. In all the results shown in this paper, the inputs have been assumed to have a linear relationship and be independent. While neural networks have been used to test the method, the results have so far been less useful than those obtained using the other methods and are not presented.

4.1 Visual fit

Simply viewing a graphical plot of the data being analysed is a surprisingly good first step in identifying relationships between data sets. This is particularly true where there is a linear and independent relationship between input and output data, and where there are relatively few data sets to be viewed. The 'analysis' process involves identifying common features (e.g. peaks) which occur regularly in both the input and output data (e.g.

rainfall and fault rates). The data can then be weighted and combined to provide a visual fit. While this approach enables hypotheses about relationships to be built, it is certainly not accurate, can be highly subjective, and therefore requires more rigorous analysis to provide more robust results.

4.2 Error minimisation

Having built a possible mathematical relationship between input and output variables, there is then a need to find an 'optimum' solution. This may be achieved by extracting an exact solution using simultaneous equations, or more likely, by minimising the sum of square errors (SSE) between predicted and actual values:

$$\text{i.e.} \quad \frac{\partial(\sum(\text{actual} - \text{predicted})^2)}{\partial(\text{input variables})} = 0$$

This solution can be found using readily available software such as Microsoft Excel's Solver, which uses numerical analysis techniques to adjust iteratively the input weightings to provide the minimum error. This method provides a quick answer, but is unable to identify relationships which exist between the input factors and to quantify the level of 'fit'. However, it can be used very effectively to assess the impact on the SSE as input variables are introduced or combined together.

4.3 Regression analysis

A more sophisticated approach to error minimisation is to apply regression analysis [7, 8]. Not only does this method provide an optimal solution by minimising the SSE as described previously, it also provides additional information on the quality of fit (coefficient of determination), on the accuracy of the weightings (standard deviations), and on any relationships between input variables (co-variance). This additional information enables error bounds (confidence limits) to be placed on the solution found (i.e. the answer is $x \pm y\%$), and helps to identify where one or more input variables are inter-related (i.e. not independent) and hence where inaccurate assumptions exist within the model. The main difficulty in this method is ensuring that the results are fully understood, and that sufficient input data is available to give a statistically meaningful result.

4.4 Neural networks

The analysis methods described so far require the user to develop the mathematical model which can then be analysed. An alternative method is to use neural networks which acts as a 'black box' approach to fit the data without

the user needing to know the weights explicitly, and which can combine inputs (see Fig 3) and use nonlinear relationships. The solutions are achieved using exactly the same approach as described for error minimisation (i.e. changing internal weightings to seek the smallest error). However, the software tools available (e.g. Matlab™) include sophisticated search algorithms which can avoid 'local minima' solutions by including concepts such as 'momentum'. While this approach is more sophisticated, it tends to require more data than regression analysis, and does not easily provide either the underlying relationships (i.e. why things happen), or the statistically useful information produced by regression analysis.

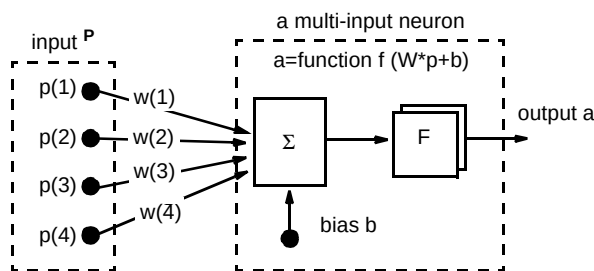


Fig 3 Simple neural network.

5. Results and discussion

5.1 Fault analysis — fault drivers

One of the first applications of the generic fault generation model was to analyse the fault generation process on a month-by-month basis to understand the behaviour of the access network in response to fault drivers. The model used is illustrated in Fig 4, and can be seen to be a sub-set of the generic model (Fig 2). To simplify the analysis, it was assumed that the sensitivity (i.e. weights) and network reliability stayed relatively constant, noting that the fault rate has not changed significantly over this period, and could therefore be ignored in the early stages of analysis.

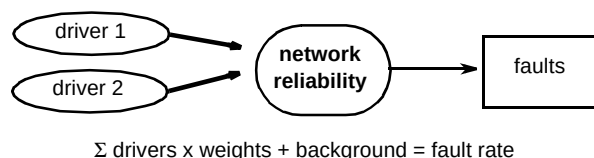


Fig 4 Fault driver model.

To analyse the system, the external inputs (drivers) were taken to be:

- manpower associated with network intervention (e.g. maintenance, repair, build) measured in man-hours using BT data,
- weather (e.g. wind, rain) using rainfall as the measure provided by the Meteorological Office,
- a background level to cover intrinsic failures (e.g. ageing, manufactured defects).

This data was taken for a 24-month period (March 1993 to February 1995) using monthly zonal data (i.e. build hours in the month, rain in the month), and is shown in graphical format in Fig 5.

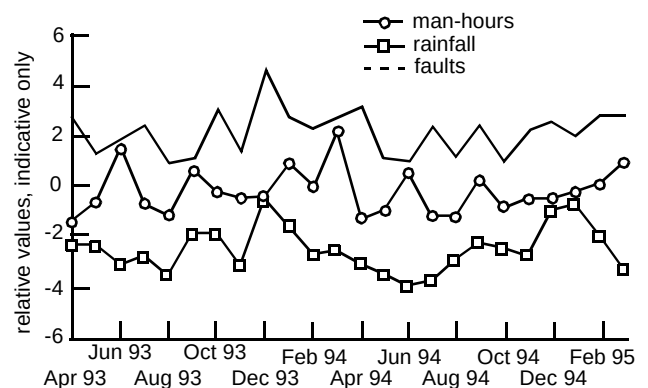


Fig 5 Raw data for fault driver model.

Within the input data (weather and man-hours), it is possible to identify visually potential 'events' which signify a possible correlation with the outputs (faults). For example, the fault peak in December 1993 coincides with a peak in the rainfall data, as does the general trends of rainfall and faults (higher in the winter, lower in the summer). In addition, numerous peaks in man-hours from the previous month coincide with those of faults in the current month (i.e. July versus June, October versus September).

The next stage, having selected potential measures which can be used to predict future behaviour, is to develop a mathematical model which links these 'inputs' (causes) to the 'outputs' (effects). In the case of this 'fault conditioners' example, the input data was correlated (fitted) with the output data using the simple equation (see also Fig 4):

$$(\text{weights} \times \text{inputs}) + \text{background} \rightarrow \text{output}$$

The weights and background were then solved to achieve the minimal error between the expected and actual outputs (i.e. between the actual and fitted fault rates). The preliminary results achieved using this straightforward approach can be seen in Fig 6 which shows the

contributions of weather, manpower, and rainfall to provide the fitted against the actual fault rate.

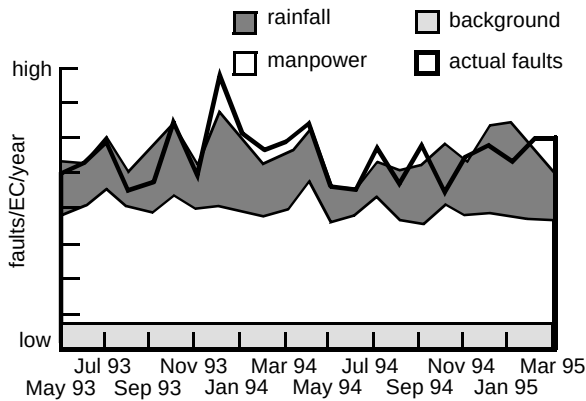


Fig 6 Actual versus fitted monthly fault rates.

The results show a number of interesting features. Firstly, the predicted fault rate underestimates the actual rate for the month of December 1993 at a time where manpower is reasonably constant and where there is a significant peak in rainfall. This mismatch is possibly due to the use of a 'linear model' which assumes that rainfall as a measure is proportional to faults. The use of a different weather measure (e.g. wind gust speed) or using a non-linear estimate (e.g. rainfall) may improve these results significantly.

Secondly, the manpower peaks after August 1994 appear to no longer lead the fault peaks by one month, seeming instead to occur in the same month as the faults. This change may well be due to a change in the systems used to record manpower effort at about this period or inconsistencies in the data itself.

Thirdly, and possibly most useful from a business viewpoint, the model indicates that faults are revealed primarily by manpower intervention and by adverse weather in approximately equal amounts, together with a smaller contribution due to intrinsic component failures.

This information, although stating what to many is 'obvious', therefore supports the view that reduced intervention in the network can potentially make a significant improvement within the fault generation process, as could improved weather-proofing of plant. These improvements could be obtained through improved tools (e.g. finding faults without intervention), processes (e.g. allowing more time to improve repair quality), or improved technology (e.g. more robust plant designs). An example where this lesson has been applied is BT's 'Optical Telecommunications Infrastructure in the Access Network' (OTIAN), which incorporates the philosophy of minimal technician intervention within the system design.

Currently, this modelling work is aiming to provide more detail (i.e. differentiate between overhead and underground fault rates), to improve the quality of the results by improving the model as discussed earlier (e.g. using nonlinear elements where appropriate), and to test the ability of the model to forecast faults by using more recent weather and manpower data.

5.2 Fault analysis — fault conditioners

A number of initiatives have been carried out within the copper access network which have had a considerable beneficial long-term impact on the rate of fault generation. These initiatives therefore present an opportunity to act as case studies to provide more specific information about the fault generation process. This is particularly true where only one or two factors have been changed over specific time periods, rather than using the more general approach which relies more heavily on derived relationships. One of these initiatives, known as 'stabilisation' [9], involved the investment of effort to pre-provide lines to customers to reduce both the lead-time on installation and to reduce the effects of random intervention as a result of line provision and installation (P&I). This case study focuses on how the network condition (in terms of fault rate) can be changed, rather than the previous analysis which concentrated on the fault generation during 'business as usual'.

In this example, two approaches were tried. The first approach used the simple model (using error minimisation) whereby faults are a function of the inputs (e.g. build, P&I) as per the previous analysis:

$$(\text{weights} \times \text{inputs}) + \text{background} \rightarrow \text{output}$$

The second approach built on the idea that effort expended would have an impact on the current fault rate due to plant being either re-conditioned or replaced as part of the stabilisation activity. In this second approach, the 'background' fault rate was therefore also changed as a result of previous build effort (the 'changing weights' line in Fig 7). The results for both these approaches are shown in Fig 7.

The predicted results correlate well with the actual fault data, with the 'changing weights' results providing a significantly lower sum of squares error (SSE). In practice, it can be seen that a closer fit has been achieved for the 'changing weights' results at the beginning of the data (i.e. April 1991—August 1991), and has reduced the amount of

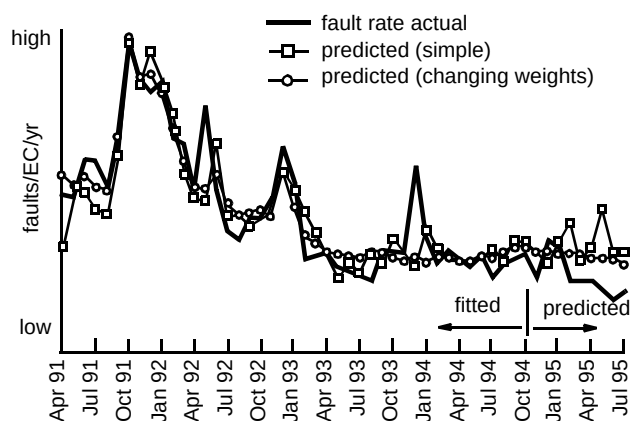


Fig 7 Predicted versus actual fault rate pre/post stabilisation (52 months).

‘overshoot’ exhibited by the simpler approach (e.g. June 1992). However, some peaks (e.g. December 1993) have not been accounted for by the model, implying other factors are present which have not been modelled.

Using the weights to predict the fault rate behaviour after December 1995 indicates that the simpler analysis yields a fault rate which varies significantly in comparison with the more complex ‘changing weights’ approach, which in turn provides the better estimate. However, in both cases the actual fault rate can be seen to have fallen still further after this date, which may provide a further opportunity to identify if other factors, such as new processes, have been introduced.

From the business view point, the data illustrates that intervention in the network appears to cause an increase in the fault report rate (i.e. pain), with a longer term reduction (i.e. gain). This ‘pain’ versus ‘gain’ can be used to provide an estimate of the long-term benefits in fault generation that can be obtained during upgrade work in addition to other gains (such as reduced intervention during P&I as in this case study). The case study therefore provides boundaries both for absolute fault rates that may occur during improvement programmes, and for fault rates which may potentially be achieved by changing the network condition.

6. Conclusions

The most useful analysis method has proved to be regression analysis which can provide confidence levels on results in addition to the most optimum fit. However, showing the data graphically has proved to be invaluable to validate results and to identify potential links between input and output data.

The most successful approach to the analysis has proved to be one of top-down refinement. This has meant starting with national data on yearly or monthly time-frames to prove general relationships, and then breaking into more

detail (i.e. by zone) and using specific case studies down to exchange level where data is available. However, practical limitations in terms of the data available (i.e. what is collected) often limits the level of network detail and data collection time-frames.

The model developed illustrates that a number of factors exist which impact on the copper access network fault rate, which include manual intervention and extreme environmental conditions. In addition, each factor has two main impacts: as a fault driver revealing weaknesses in the network, and as a network conditioner changing the reliability (i.e. age and condition) profile. While this model has only been applied to copper access networks, it is also likely to hold for technologies such as fibre, with the lessons learnt having already been incorporated within BT’s OTIAN design.

Since intervention and environment appear to be major fault drivers, this implies that the reverse is true, i.e. minimising intervention and ‘weatherproofing’ the network is likely to yield long-term benefits in terms of reliability. However, any changes to processes, tools or technology are likely to cause faults initially before long-term benefits can be seen. The challenge is therefore to introduce programmes which make improvements without introducing any significant and unwanted side-effects.

Acknowledgements

The authors wish to thank all those who have helped to provide the data, and, in particular, those involved with the access operation unit [9], Clydenet, and the fault volume reduction programme office — without this help, no analysis would have been possible.

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